

SOP-4412 Vessel Shutdown and Depressurisation Procedure

1. Purpose

This procedure defines the controlled shutdown and depressurisation of pressure vessel V-1201 in the Crude Distillation Unit. It applies to planned turnaround shutdowns and to controlled emergency depressurisation.

2. Scope

Applies to V-1201 and its associated relief system PSV-1207. Does not cover emergency blowdown initiated by the Safety Instrumented System.

3. Responsibilities

The Shift Superintendent authorises the shutdown. The Panel Operator executes the depressurisation sequence. The Inspection Engineer confirms that isolation is complete before any entry permit is issued.

5. Shutdown Sequence

5.1 Feed Isolation

Close feed valve HV-1205 and confirm zero flow on FI-1205. Isolate the vessel from the flare header by closing XV-1207. Both valves shall be car-sealed closed and tagged before proceeding.

5.2 Depressurisation

Depressurise through the 2-inch vent to flare. The depressurisation rate shall not exceed 2 bar per minute to avoid thermal shock to the shell. Hold at 4 barg for four hours at 120 degC to allow residual hydrocarbon to vaporise before final venting to atmosphere.

5.3 Nitrogen Purge

Purge with nitrogen until the hydrocarbon content measured at the vent is below 1% LEL. A minimum of three volume changes is required. Record the final gas test on Form OISD-105.

6. Safety Requirements

Confined space entry requires a valid permit, continuous atmospheric monitoring, and a standby attendant. Refer to OISD-STD-105 for hot work and IS 3844 for fire safety provisions.

7. Retirement Criteria

The minimum required shell thickness for V-1201 is 8.0 mm as calculated per ASME Section VIII Division 1 UG-27. Any measurement at or below this value requires immediate fitness-for-service assessment before the vessel is returned to service.